Lab 16 — Deploy a .NET Framework App on Windows Pods with gMSA + IIS

K8S-LAB-16 · Windows · Enterprise · 120 min

Focus: IIS, .NET Framework, gMSA, Active Directory, Windows secrets

Objective

Containerize a legacy .NET Framework app on IIS, run it on a Windows node, and authenticate to SQL Server using a Group Managed Service Account.

Concepts Covered

• Group Managed Service Accounts and the GMSACredentialSpec CRD

• IIS as a container entrypoint (ServiceMonitor.exe)

• Why .NET Framework must use Windows containers

• RBAC binding the ServiceAccount to a credspec

Prerequisites

• Hybrid cluster from Lab 11

• AD domain reachable from Windows nodes

• AD admin to create gMSA + KDS root key

Architecture

```
Active Directory <--Kerberos--+
                              |
Windows worker (domain-joined)
   kubelet -> containerd-windows
        |
        +-- IIS Pod (gMSA: webapp$)
                |
                v
       SQL Server (Windows Auth, no password)
```

Lab Steps

01. Create the gMSA in AD

```
Add-KdsRootKey -EffectiveTime ((Get-Date).AddHours(-10))
New-ADGroup -Name WebAppHosts -GroupScope Global
Add-ADGroupMember WebAppHosts -Members WIN-NODE-1$
New-ADServiceAccount -Name webapp -DNSHostName webapp.corp.local -PrincipalsAllowedToRetrieveManagedPasswor
```

Expected: gMSA webapp$ created.

02. Install GMSACredentialSpec CRD & webhook

```
kubectl apply -f .../gmsa-crd.yml
kubectl apply -f .../gmsa-webhook.yml
```

Expected: CRD gmsacredentialspecs.windows.k8s.io established.

03. Generate and register the credential spec

```
Install-Module CredentialSpec -Force
New-CredentialSpec -AccountName webapp
kubectl apply -f webapp-credspec.yaml
```

Expected: gmsacredentialspec/webapp created.

04. RBAC: allow ServiceAccount to use credspec

```
ClusterRole(use on gmsacredentialspecs/webapp) + RoleBinding to default SA
```

Expected: Role + binding created.

05. Deploy the IIS app referencing the gMSA

```
Deployment spec includes:
  securityContext:
    windowsOptions:
      gmsaCredentialSpecName: webapp
```

Expected: Pods Running on Windows node.

06. Validate SQL Integrated Auth

```
kubectl exec -it deploy/legacy-webapp -- powershell -c "Invoke-Sqlcmd -ServerInstance sql.corp.local -Query
```

Expected: CORP\webapp$

Common Errors

failed to create credential spec
Cause: Node not in AD group allowed to retrieve gMSA password
Fix: Add node computer account to WebAppHosts and reboot

Pod rejected by webhook
Cause: ServiceAccount lacks 'use' on credspec
Fix: Apply ClusterRole + RoleBinding from step 4

SQL auth fails as ANONYMOUS LOGON
Cause: Missing SPN on SQL service
Fix: Register SPN MSSQLSvc/sql.corp.local:1433

Troubleshooting

• Inside Pod: nltest /sc_verify:corp.local

• Inside Pod: klist

• On node: Get-ADServiceAccount webapp -Properties *

Enterprise Relevance

gMSA is the only supported way to give containers Windows identities — every Windows-on-Kubernetes adoption hits this lab eventually.

Validation

• Pod runs on Windows node

• klist inside Pod shows webapp$ TGT

• App queries SQL with Integrated Auth, no secrets in manifests

Questions

[Beginner] Q: Why can't .NET Framework run in a Linux container?

A: It requires the Windows API; only .NET Core/5+ is cross-platform.

[Intermediate] Q: What does gMSA solve?

A: Automatic password rotation in AD; multiple hosts share one identity.

[Advanced] Q: Where is the gMSA password stored on the node?

A: Nowhere — node fetches on demand from AD using its machine account.

[Scenario] Q: SQL logs 'Login failed for CORP\WIN-NODE-1$'. Cause?

A: gMSA spec not applied; node identity leaked. Re-check gmsaCredentialSpecName.

[Viva] Q: Can the same gMSA be used by Pods on multiple nodes?

A: Yes — add every node's computer account to the AD group.